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Nueralink: Uploading the Brain

Explore #2

2a EFFECTS:

Elon Musk's Nueralink has many pros and benefits to society as it does disadvantages. Some of those benefits it can bring to our society include being able to operate computer with our brain. This will allow patients with lost limbs or those who are paralyzed to once again be able to move their body parts. Not only will help physical ailments but the Nueralink is also designed to help those who are suffering from neurological disabilities as well. As technology advances the Nueralink could also one day replace our need for smartphones, tablets, and computers because the Nueralink would be able to provide all that information directly to our brain. All these advantages have their adverse as well, one disadvantage might be that they are susceptible to hacking. If these chips are that hacked that are implanted into the brain, then any one hacker would be able to gather all the information the Nueralink collects on the brain. The hacker could possibly as well if advanced enough be able to take control of those synthetic limbs and control them remotely, or even shutting off the ability the person has to control that limb or any limb. Another doubled edged sword that the Nueralink has is cybernetic enhancements. Cybernetic enhancements could be cool in theory but in practicality can open a whole new world of dangers. This could lead to cyber enhanced acts of terrorism or even cyber enhanced warfare between nations. One day our society could be living in the world of Cyberpunk. Another danger that is hypothesized is that with the Nueralink it would be possible to implant false memories or even control a victim's thoughts. All the advantages and disadvantages are just possibilities, and it will ultimately come down to how we the user intend on using the capabilities of the Nueralink.

2b DATA:

The Nueralink captures data by having a chip placed inside a small section of your brain. The chip will have small electrodes that are pushed and scattered throughout the brain. There will be approximately three thousand electrodes scattered throughout analyzing, interpreting, and collecting the brains signals. These electrodes will primarily be placed on the main parts of the brain including, the visual cortex, auditory cortex, somatosensory cortex, and motor cortex. The electrodes will be light and thin which allows for 360-degree rotation. When the electrodes are in place, they will be able to analyze and detect the communication between neurons. After the electrodes will send these signals back to the chip where the chip is able to convert these signals into binary language so other computers can interpret it and be able to run algorithms on those signals. Each chip is limited to recording only about one thousand neuron signals this limitation is a big concern for data storage. The Nueralink receives its input from the brain and the electrical signal that are passed between the different neurons inside. The Nueralink outputs its data to your smartphone using Bluetooth and other various forms of wireless technology. This allows any user to be able to see their own brain activity. The app will also provide lessons for the user to teach them how to use the Nueralink and get comfortable controlling electronic devices with the brain.

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2c CONCERNS:

One major data storage issue the Nueralink has is that a single chip can only process up to one thousand neural signals at a time. This limitation makes it so that there is not enough data being captured to make sense of what the brain is trying to accomplish. This means that users will have to install multiple Nueralink chips around four to ten in their brain for enough neural signals to be processed by each chip. This leads to another data storage concern and that your brain is always processing information and so how will all this data be stored over extended periods of time because it will begin to pile. Finally, the Nueralink also has a data security concern and that is if the device gets hacked. If the Nueralink where to be hacked then you could see all the information going on the person's brain that is a security vulnerability because then big figure heads who have this technology could be listened into, their secrets revealed. This would be a massive disadvantage to privacy and what personal information.

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